Amrit Sanjeev | Aerospace Engineering Student

Twerton, Bath, United Kingdom

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Education

University of Bath, UK, BEng Aerospace Engineering (Sep 2025 - June 2029)

Some 1st-year modules:

- Engineering Mathematics (ME12001): 68%
- Thermofluids (ME12003): 70%

Ressun Lukio, Finland, International Baccalaureate (IB) (Aug 2022 - May 2025)

- Pure Mathematics Higher Level: 7
- Physics Higher Level: 7
- Chemistry Higher Level: 6

Delhi Public School BN, India, CBSE (2013 - 2022)

Final 3 years (Notable grades): 3 A1s, 4 A2s, 4 B1s

Engineering Projects & Experience

- **Ping-Pong Ball Launcher (University Collaborative Project)**

As part of the project, I had to construct a mechanism involving two horizontal flywheels for propulsion and enabling control over the yaw and pitch of the projectile's direction. My role involved designing most of the complex parts using CAD and also computationally simulating the flywheel and steering mechanisms.

This project helped me develop various skills such as: Collaboration on large, complicated projects. Problem-solving to design components that fulfil the desired functionality. Organisation & delegation of tasks for better efficiency, and also optimisation & refinement of mechanisms and ideas via software-human feedback loops.

- **Investigating Aerofoils (IB Extended Essay)**

As part of my Physics Extended Essay (3000 words) during the IB, I investigated the effects of applying loads on a flexible (homogeneous) aerofoil to simulate the stresses and resultant strain it would experience in actual flight. It forced me to develop independent & efficient learning skills.

Computing Projects and Skills

- **Machine Learning (Rust and OpenCL)**

A multi-layer perceptron that tries to predict and classify handwritten digits from 0 to 9. The model was trained on MNIST's database using Rust and hardware-accelerated using OpenCL.

- **Fluid Simulation (C++ and OpenGL)**

A 3D fluid simulator that uses a grid solver to simulate fluid parcels and calculates various properties of the fluid within those parcels, such as velocity, pressure, temperature, etc. The simulator allowed multiple custom 3D mesh inputs and simulated fluid flow around them.

- **Python**: Fetched & parsed ERA5 temperature data presented on a shader-generated globe.
- **Autodesk Inventor**: Used CAD to design and model complex 3D components.
- **MATLAB**: Simulated and recorded sensor control data loops; data processing.
- **Claude Code**: Leveraged custom MCP connectors and skills to optimise problem-solving.

Work Experience

I worked as a Summer Trainee at Nokia, Espoo (Jun 2026 - Sep 2026) in the Radio Layer (RL) - 1 troubleshooting department. During which I gained significant experience in how telecommunications work at its most fundamental level. My role involved debugging, refactoring and implementing new features based on bug reports submitted by people within the company or externally. Most of the internship was a journey into unfamiliar domain concepts, forcing me to learn nearly everything on the fly, aided only by my programming skills.

One of the notable projects I contributed to involved restructuring the mechanism used to collect telemetry data across the fronthaul. I redesigned the software so that a single request from the upper command layer could trigger data collection across both the uplink & downlink pathways. This required me to understand how the various components of the system interact, how data flows between them, and how these processes are controlled through software.

Activities and Interests

Tennis

Tennis has always been one of my favourite sports. I have competed and even won local tennis tournaments near my home. Tennis requires quick thinking and intense focus. As such, playing tennis always helps me clear my mind.

Video Gaming

Video Gaming is one of those hobbies that requires a vast set of skills. E.g., CS: GO requires both quick reflexes and long-term strategic thinking, allowing for a challenge while being fun.